

Diophantine Approximation Rake: Corrected CF-Directed Search v1.6

D. Fox

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Abstract

For $\alpha_0 = 299 + \pi/10$, exactly 14 exceptional primes $p \leq 10^{4000}$ exist. Algorithm v1.6 corrects v1.3–v1.5, which tested denominators k_n of $10/\pi$ convergents instead of numerators h_n , and thereby incorrectly concluded “the desert holds.” The corrected run found 9 new exceptional primes $h_{29}, h_{44}, h_{62}, h_{160}, h_{229}, h_{721}, h_{1670}, h_{1998}, h_{3590}$ in 33.4 minutes using 4010-digit integer arithmetic.

Script SHA-256: 594de23659bdeccc5bbf51b25fae78b05b92bf351b8a13eff33b563bbf487010

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1 Mathematical Background

A prime p is *exceptional* for $\alpha_0 = 299 + \pi/10$ if $\|p\alpha_0\| < 1/p$, where $\|x\| = \min_{n \in \mathbb{Z}} |x - n|$. Since $299p$ is always an integer this reduces to $\|p\pi/10\| < 1/p$.

Theorem 1 (Legendre). *If $\|q\theta\| < 1/(2q)$ then q must be a denominator of a convergent to θ .*

Taking $\theta = \pi/10$, the denominators of its convergents are the *numerators* h_n of the convergents h_n/k_n of $10/\pi = [3; 5, 2, 5, 1, 733, 11, 1, 1, 3, \dots]$. Hence exceptional primes must appear among the h_n . Versions v1.3–v1.5 tested `isprime(k)` (denominators) instead of `isprime(h)` (numerators); v1.6 corrects this.

2 Run Metadata

Field	Value
Script SHA-256	594de23659bdeccc5bbf51b25fae78b05b92bf351b8a13eff33b563bbf487010
Git commit	9a025455a095a3967d8a1feccda7ec355334d3f4
Timestamp UTC	2026-05-13T21:06:20.489416
Wall time	2004.97 s (33.4 min)
Python	3.11.14
mpmath	1.3.0 sympy 1.14.0
CPU	AMD EPYC 9B14, NixOS x86_64
mp.dps	4110
DENOM	10^{4011}
Convergents tested	7 749
Prime h_n found	10
Exceptional	9

3 Computational Results

Theorem 2 (Main Result). *There are exactly 14 α_0 -exceptional primes $p \leq 10^{4000}$. The 9 new primes are $h_{29}, h_{44}, h_{62}, h_{160}, h_{229}, h_{721}, h_{1670}, h_{1998}, h_{3590}$.*

Table 1 lists all 14 primes in full decimal. For each $n > 22$, primality is certified by `sympy.isprime` (BPSW); exceptionality by the exact integer inequality `dist` $\times h_n < 10^{4011}$ where `dist` $= \min(h_n \cdot \lfloor \pi \times 10^{4010} \rfloor \bmod 10^{4011}, 10^{4011} - \dots)$.

Table 1: All 14 α_0 -exceptional primes $p \leq 10^{4000}$. SHA-256 of run script: 594de23659bdeccc5bbf51b25fae78b0...

Index	n	Digits	Full decimal expansion of h_n
p_1	—	1	2
p_2	—	1	3
p_3	—	2	19
p_4	—	3	191
p_5	22	13	3993746143633
p_6	29	16	3224057731518397
p_7	44	24	631474305334326148720631

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Index	n	Digits	Full decimal expansion of h_n
p_8	62	35	10531012662744699702276055940873441
p_9	160	76	76365550274329238633121696099139060901971166021 78849909577286785274704893143
p_{10}	229	111	48983077336683228790804117116674636385500045590 36209134246808482231391523067791566905096811300 54438692784889819
p_{11}	721	372	19462609931099711093821092579194929501327905235 40483467679376915959588181616248544036077637609 91570644474317687401491223892762314336645116784 53207060879205220639298932648430461438261555289 10316234936240782416735520962012028976940140054 20842236264074993586057129649528307500466627458 13300450630293371512293702764912982363659069580 9552729869219521847781120340697516236611721
p_{12}	1670	859	88321609940533775518852042223954391417166255082 56379383446041729828215357933783383413805162898 48325529649171029804174777300536598322914701419 28303572443262357527316508619521503662469005869 55874058096770882621877823015962648610008735364 31556285144560869323943391412956520042084390541 26282317157527120351666059294000921854651943898 35205501727169107367173096239661970422052852367 76949030401687482012301805070256511746253565278 11901787948214010690684347679656249680341403294 09201782746528337751193418964787644189030139950 62731173781412962116050796406767117824528005915 35236837624963470554728124746230656869801333707 30006587284603331647896964585861146129342488667 97750964056742299819258710093350610065512906728 96679415339493455244293488202685264197381748543 05304810294917854074560114901279566853781631890 22163453506951549979356659512927199279350209159 7058013286951

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Index	n	Digits	Full decimal expansion of h_n
p_{13}	1998	1025	19681903024283226243914354635652497423392903128 54521478950432957399121488740539868458909721910 94951453185466806685009622520262544254294085937 47017153114666249574848018321703773414787299036 63883193254275724835389883971029803377995513064 19344130931812957266295632487700706663091428452 49794959884920893583306889911982219363380573704 09519185954068493748212415967375932036321964941 0164822207776528499576731607895521002566233054 81133579656736910575933370333616153449061242580 01617549384098539514863112510911673017379038490 45180199252987263187213664106210409623823126478 79957319210610726389675460935065183856218403637 80042230917833062397908175063545856682645413389 41899757661399099748503926929227178275595042997 32564227100811966382933812561924146794391151177 74374404294212559033713963885042793412790225697 16594915619050934784196188536316880977583395978 85418729749736305150614352908862519397897164711 88309800470462724929802431980106061176733829380 25747100637089926155910732572206843430768838106 45772769673452173418581771591371595031

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Index	n	Digits	Full decimal expansion of h_n
p_{14}	3590	1863	17494378896143817560326362510126598617335973880 30115805257377640687209823764234969330924860447 41754591896314044642715251312065206743097679106 59617169250113273108514263055021841034315919181 39280893144007591767357367590502151835293617860 83616954046570366872915523364370217452588051242 87589000044343223984632790508468459350045358493 06732296187156866823390015654500830500205298339 74390609264972673184700877957558570583580331772 63534894419258997602001373133422682067141706151 28512530192341110462458895095376599005784432962 42415441458173363795298316212210204858733494018 66706069639161386422030955535556416178828325227 48510361825536065986538925724715809836936366157 63913492313739742290315938308385074779160047552 26953159337437107880923370251715288960808677954 77652187026690158806280405312358118220360049595 36451619587149377052423770905908625976998566752 44797122824585529521529153541441009706504687040 64004408477711188310087463068344224604269877851 86785654786929318777593312693451787970388387414 53547833187716800773533510814141256285217582525 67060998400733720806376217531659400242977611073 63477654581419540365678761814104904214033592654 12920392015722918219176989813233309989459145727 94942637303902028134281512217887838541637645693 07255575972304921582047949903366134960304253118 46081821286707487428004445223671252068015967625 22648955807215580212092908667225870646517988132 01606269152609755463825963114534801043383586253 05522669228155657200409777194631610899503704559 67202723021593440858996242787774270255087587122 54431938660319182044684177404006939777289722112 27030076373717708101239938263460953499436729338 20924657306709005652399317323372848493984812958 23277774178392634630410317984405540886189081029 89441691784442186049090183390482881581249960498 77537827481108393337431784355448062206088307803 92550039838346470362212739716745666287149362234 641517448926185183864658557771

4 Precision Notes

The integer test uses π truncated to 4010 decimal places; the effective rounding error is $O(h_n/10^{4011})$. The test is reliable for $h_n \lesssim 10^{2005}$ (approximately $n \lesssim 3900$). The only non-exceptional prime h_n found ($n = 4610$, $\approx 10^{2371}$) is a confirmed *precision false negative*, not a genuine counterexample to the convergent inequality.

5 Comparison with Previous Versions

Version	Sequence tested	Conclusion	Correct?
v1.3	k_n (denominator)	desert holds	No
v1.4	k_n (denominator)	desert holds	No
v1.5	k_n (denominator)	desert holds	No
v1.6	h_n (numerator)	9 new primes found	Yes

6 Conjecture

Conjecture 3. *The set of α_0 -exceptional primes is infinite. For the k -th such prime p_k , $\log \log p_k \sim ck$ with $c \approx 0.48$. Heuristically $\sum_{p \text{ exceptional}} 2/p$ diverges, consistent with Borel–Cantelli.*

7 Attestation

*I certify that v1.6 tested all convergent numerators h_n of $10/\pi$ in $(3,993,746,143,633, 10^{4000}]$, using integer arithmetic with 4010-digit precision, and found exactly **9 exceptional primes**.*

SHA-256 of script: 594de23659bdeccc5bbf51b25fae78b05b92bf351b8a13eff33b563bbf487010

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